Review of Bayesian Regression

EES 4891-06/5891-01

Bayesian Statistical Methods

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Linear Regression

Linear Regression

- Start with simple regression:
  - One predictor variable X
  - Predictor and outcome are both continuous (real numbers)

$$Y \sim \text{Normal}(\mu, \sigma)$$
$$\mu = \alpha + \beta X$$

- Multiple linear regression
  - N predictor variables $X_1, \ldots, X_N$
  - Everything continuous

$$Y \sim \text{Normal}(\mu, \sigma)$$
$$\mu = \alpha + \sum_{i=1}^{N} \beta_i X_i$$

- Polynomial regression
  - Just treat each power of X like a new predictor
  - You can have polynomials with more than one variable

$$Y \sim \text{Normal}(\mu, \sigma)$$
$$\mu = \alpha + \sum_{j=1}^{N} \beta_j X^j$$

Scaling Variables

- Standardizing:

$$X_{\text{std}} = \frac{X - \bar{X}}{\sigma_X}$$

  - All variables on the same scale
  - Centered with 0 at the mean
  - Slopes (β) measure the effect of changing by 1 standard deviation.
  - If it's important that $X \geq 0$,

$$X_{\text{std}} = \frac{X}{X_{\text{max}}}$$

    - X_{std} ranges from 0 to 1.

- Log scaling
  - Good for outcome variables that **must** be positive
  - Good for predictor variables with a tail that covers a large range
    - Population is often log-scaled
  - Variable must be > 0, so if some values are 0, add a small number to them (e.g., 0.01, 0.001).

Integer Models

- Boolean (Yes/No)
  - Coin toss: Heads or tails
  - Bernoulli distribution
  - Special case of binomial, with one trial: `dbinom(1, p)`
- Count data:
  - Binomial distribution
    - `rbinom(N, p)`
    - Maximum value = N
  - Poisson distribution
    - `dpois(lambda)`
    - No maximum value
    - Poisson(λ) is the limiting case of Binomial(N, p) when N is large and p is small, with $Np = \lambda$.

- Generalized Linear Models (GLMs):
  - Link functions:

$$Y \sim \text{Binomial}(N, p)$$
$$p = \text{logit}^{-1}(\alpha + \beta X)$$

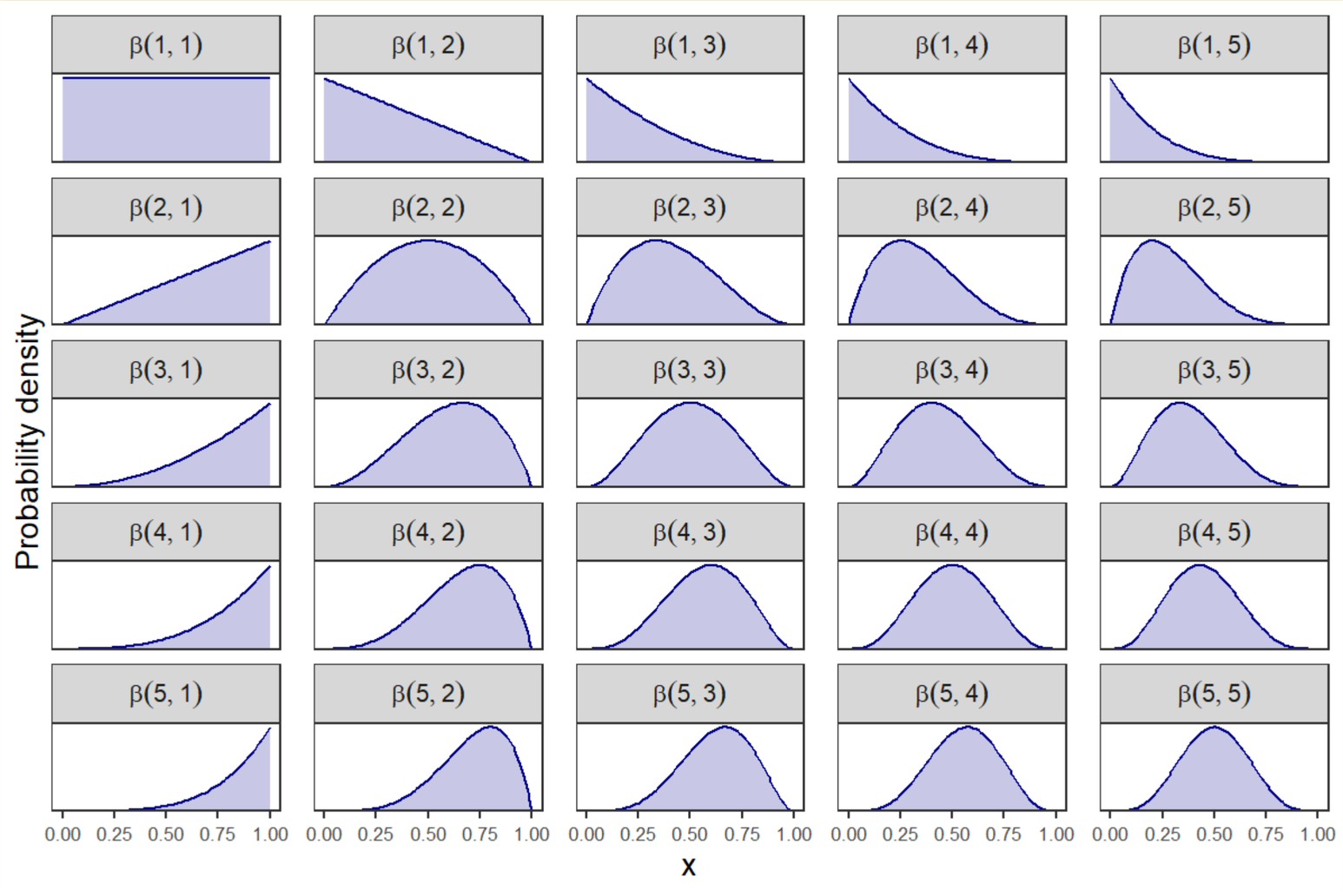
```
y ~ dbinom(N, lambda),
logit(p) = a + b * x
```

$$Y \sim \text{Poisson}(\lambda)$$
$$\lambda = \exp(\alpha + \beta X)$$

```
y ~ dpois(lambda),
log(lambda) = a + b * x
```

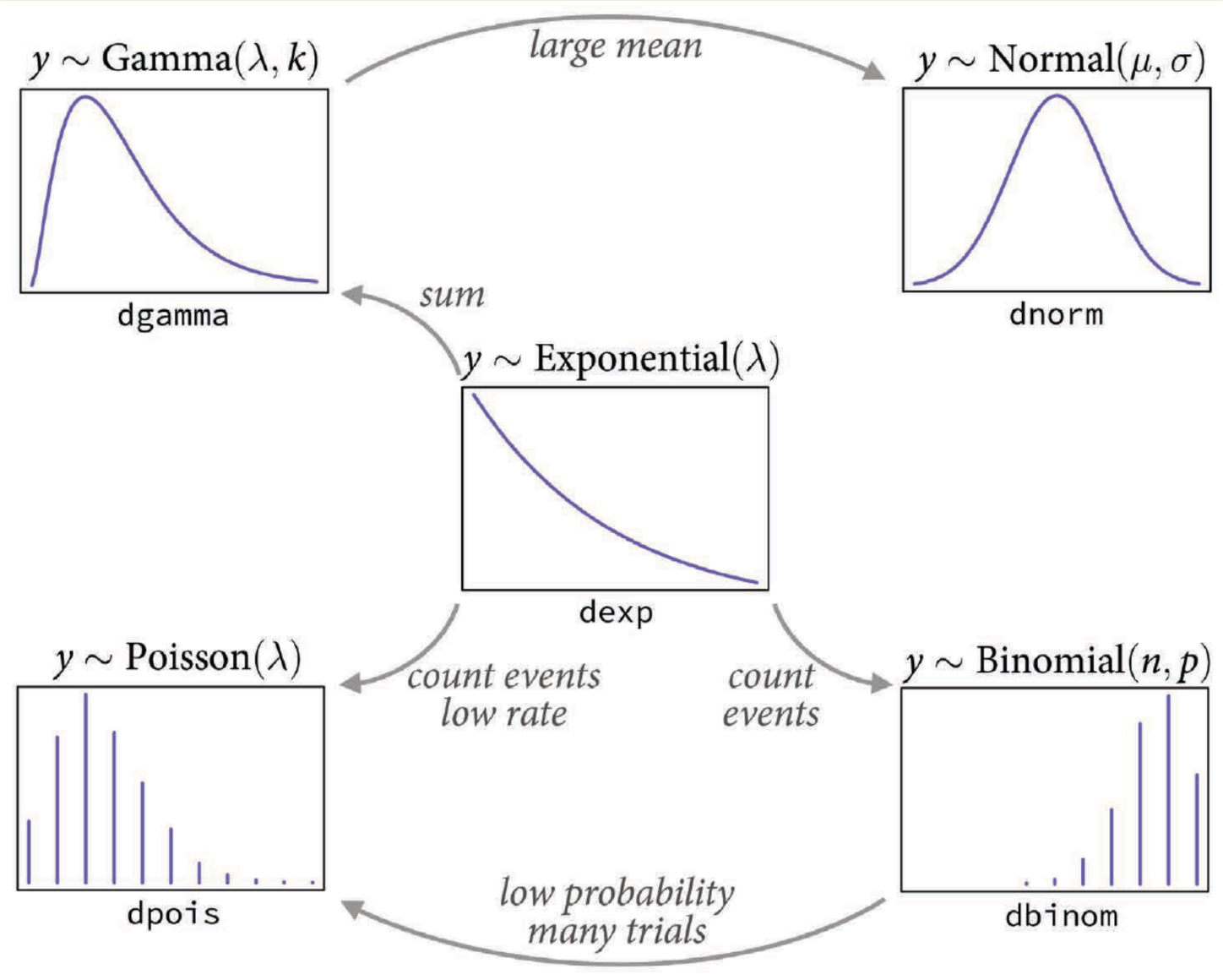
  - Priors for probabilities (range 0 to 1)
    - Beta functions: `dbeta(a, b)`
      - Mean = $a/(a+b)$

Beta Distributions



- Mean = $a/(a+b)$
- Variance = $ab/((a+b)^2(a+b+1))$

Exponential Family of Distributions



Categorical Variables

- Categorical variables
  - Predictors (section 5.3):
    - Use index variables (0 or 1)
    - N levels: $N - 1$ index-variables (default category)
      - Only one index is 1 (the one for the category)
      - If they're all zero, it's the default category
  - Outcomes (section 11.3):
    - Integer 0...N
    - Use multinomial (categorical) likelihood (`categorical()`)
    - Use Dirichlet priors (`dirichlet()`)
      - Dirichlet is like an N-dimensional generalization of the beta distribution
- Ordered categorical variables
  - Effects are cumulative
  - Predictors (section 12.4):
    - Use an integer variable (K) with values $0 \ldots N - 1$
    - $N - 1$ variables δ_i ($i \in [1, \ldots, N - 1]$)
      - $\mu = \alpha + \sum_{i=1}^{K} \delta_i$
    - Initially, no effect.
    - Every time you step to the next level, there's another effect, and they add up.
  - Outcomes (section 12.3):
    - Integer 0...N
    - Use multiple logistic regression
      - Initially at lowest value
      - As effect grows, step through values in sequence.

Multilevel Models

Multilevel Models

- Data is grouped into *clusters*
  - Geographical groupings (states, counties, etc.)
  - Temporal groupings (seasons)
  - Other categories:
    - Gender
    - Education
    - Profession
    - Species
    - Individual
    - ...
- Hyperpriors and hyperparameters:
  - Each group may have its own prior for slope, intercept, etc.
    - The parameters for that prior are drawn from a *hyperprior*

- Single-level model

$$
\begin{aligned}
Y &\sim \text{Normal}(\mu, \sigma) \\
\mu &= \alpha + \beta X \\
\alpha &\sim \text{Normal}(0, 1) \\
\beta &\sim \text{Normal}(0, 1) \\
\sigma &\sim \text{Exponential}(1)
\end{aligned}
$$

- Two-level model (varying intercept)

$$
\begin{aligned}
Y &\sim \text{Normal}(\mu, \sigma) \\
\mu &= \alpha + \beta X \\
\alpha &\sim \text{Normal}(\bar{\alpha}, \sigma_\alpha) \\
\beta &\sim \text{Normal}(0, 1) \\
\bar{\alpha} &\sim \text{Normal}(0, 1) \\
\sigma &\sim \text{Exponential}(1) \\
\sigma_\alpha &\sim \text{Exponential}(1)
\end{aligned}
$$

Multilevel Models

- Two-level model (varying intercept, varying slope)

$$
\begin{aligned}
Y &\sim \text{Normal}(\mu, \sigma) \\
\mu &= \alpha + \beta X \\
\alpha &\sim \text{Normal}(\bar{\alpha}, \sigma_\alpha) \\
\beta &\sim \text{Normal}(\bar{\beta}, \sigma_\beta) \\
\bar{\alpha} &\sim \text{Normal}(0, 1) \\
\bar{\beta} &\sim \text{Normal}(0, 1) \\
\sigma &\sim \text{Exponential}(1) \\
\sigma_\alpha &\sim \text{Exponential}(1) \\
\sigma_\beta &\sim \text{Exponential}(1)
\end{aligned}
$$

- Two-level model (correlations between slope and intercept)

$$
\begin{aligned}
Y &\sim \text{Normal}(\mu, \sigma) \\
\mu &= \alpha + \beta X \\
\begin{bmatrix} \alpha \\ \beta \end{bmatrix} &\sim \text{MVNormal}\left(\begin{bmatrix} \bar{\alpha} \\ \bar{\beta} \end{bmatrix}, S \right) \\
S &\sim \begin{pmatrix} \sigma_\alpha & 0 \\ 0 & \sigma_\beta \end{pmatrix} R \begin{pmatrix} \sigma_\alpha & 0 \\ 0 & \sigma_\beta \end{pmatrix} \\
\bar{\alpha} &\sim \text{Normal}(0, 1) \\
\bar{\beta} &\sim \text{Normal}(0, 1) \\
\sigma &\sim \text{Exponential}(1) \\
\sigma_\alpha &\sim \text{Exponential}(1) \\
\sigma_\beta &\sim \text{Exponential}(1) \\
R &\sim \text{LKJCorr}(2)
\end{aligned}
$$

Grouping by Continuous Variables: Gaussian Processes

- The previous examples divided data into groups using categorical variables.
  - Treatment condition
  - States
  - Counties
  - Clade
  - ...
- What happens if we want to group by a continuous variable?
  - Height
  - Weight
  - Distance
  - Age
  - Time
  - ...
- Gaussian Processes
  - Grouping variable is continuous (e.g., t for time)
    - Observations at $t_1, t_2, \ldots, t_N$
    - Measurements of x_i, y_i at t_i

$$
\begin{aligned}
y_i &\sim \text{Normal}(\mu_i, \sigma_i) \\
\mu_i &= \alpha_i + \beta_i x_i \\
\text{Cov}(\mu_i, \mu_j) &= K(t_i - t_j)
\end{aligned}
$$

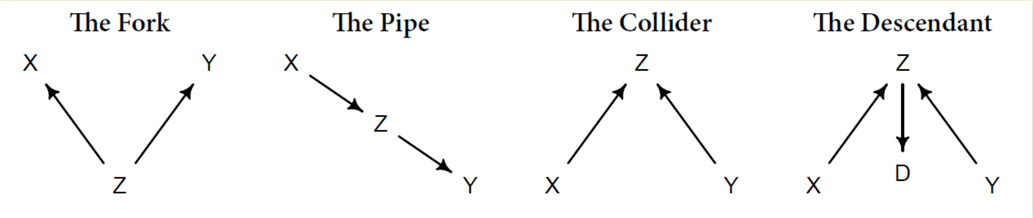
where K is a *kernel* function that describes how quickly the similarity of the relationship between x and y falls off as the time between observations grows.

- More partial pooling among closely spaced measurements.

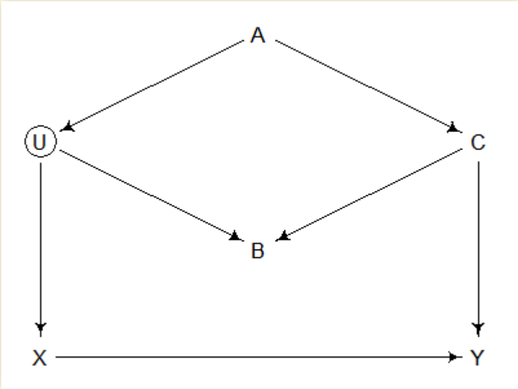
Designing Models

Designing Models

- Analyze relationships between variables
  - DAGs describe *causal relationships* among variables.
  - They can help us detect potential problems:
    - Spurious associations
    - Masked relationships
    - Multicollinearity
    - Post-treatment bias
    - Confounders
- Four fundamental types of confounding relationships:



- General rules:
  1. List all paths connecting *X* (potential cause) to *Y* (outcome)
  2. Classify each path as *open* or *closed*
     - A path is *open* unless it contains a *collider*
  3. Classify each path by whether it's a *backdoor path*.
     - A *backdoor path* has an arrow pointing at *X*
  4. If there are any *open backdoor paths*, try to close it by *conditioning* on a variable.
- Example (Section 6.4.2)



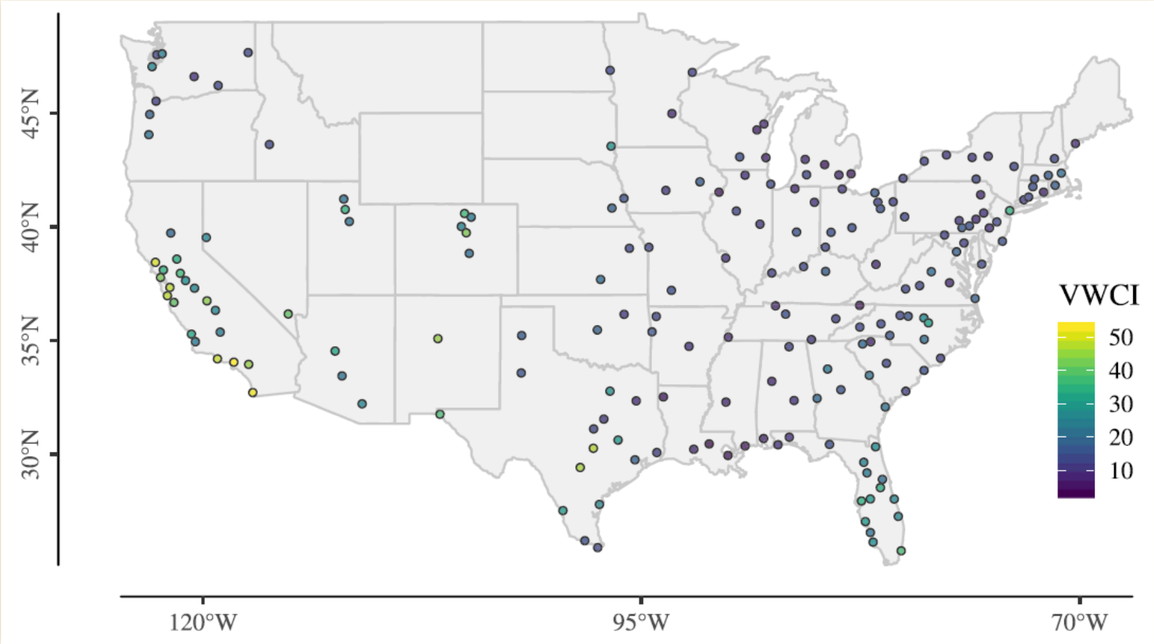
Testing DAGs

- How can you tell whether your DAG is correct?
  - You can't
  - There could always be important unobserved variables you don't know about
- Comparing DAGs with evidence
  - Analyze DAGs for *conditional independencies*
    - *Conditional independencies* are empirically testable statements
    - Use your model's posterior predictions to test predicted *conditional independencies*.
    - If DAG's *conditional independencies* are not observed, then it's probably not correct.
    - `dagitty`'s `impliedConditionalIndependencies()` function is your friend.
- DAG analysis is helpful, but it's not enough.
  - DAGs tell you about the *logical structure* of your model, but they don't tell you about the science.
  - What you know as a scientist is even more important:
    - *Why* do you think *X* influences *Y*?
    - *What* other variables might also play a role?

Example: Urban Water Conservation Policies

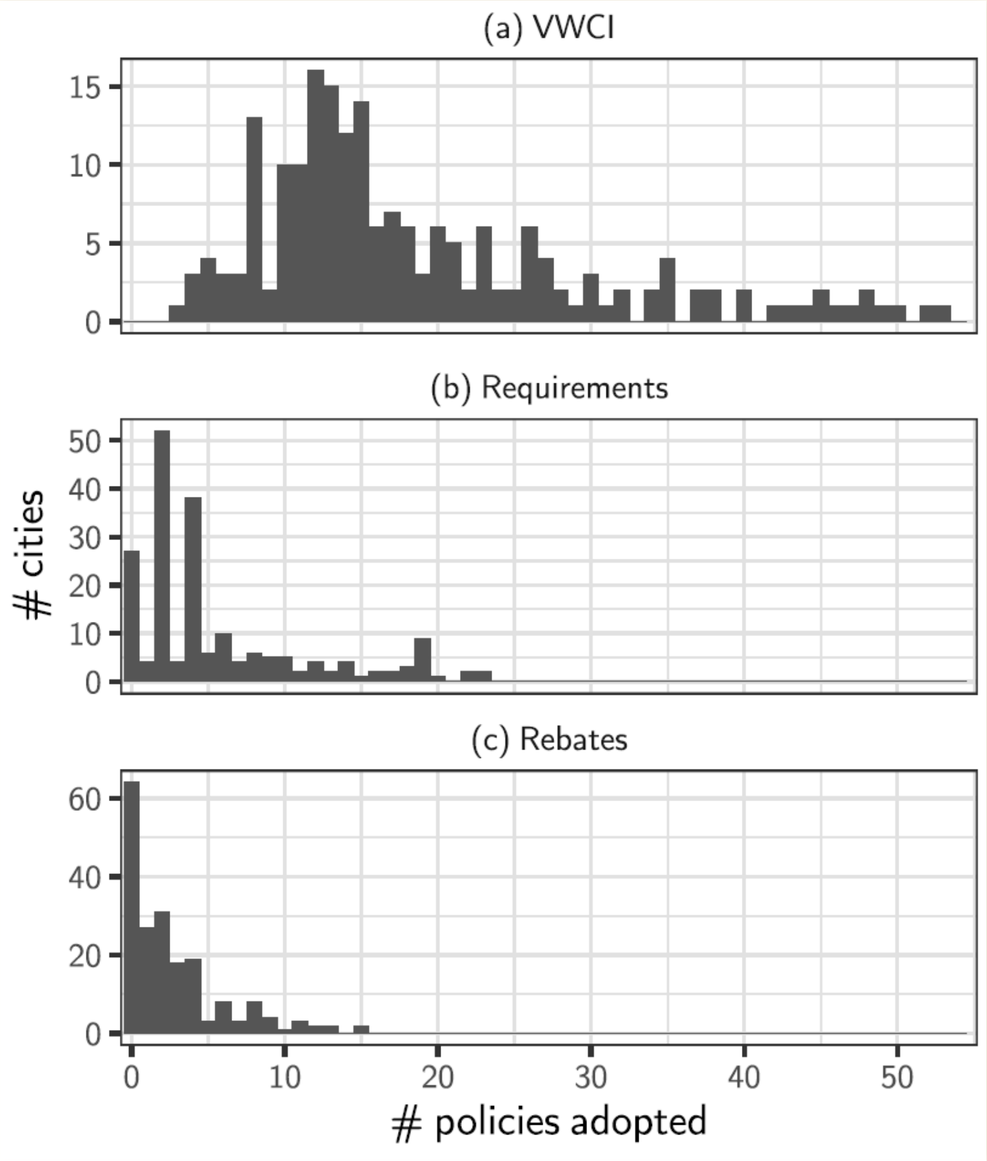
Urban Water Conservation Policies

- Build a database of water conservation policies for 197 largest cities in the US
  - Vanderbilt Water Conservation Index (VWCI)
    - List of 79 possible policies
      - 31 are *requirements*
      - 21 are *rebates* for voluntary actions
    - Each city gets a score based on how many policies it has adopted



Descxriptive Statistics

- Range of values: 3–53 (max. possible is 79)
- Mean: 18.7, median 15
- Model VWCI:
  - Predict score from
    - Temperature
    - Rainfall
    - Fraction of water supply from surface water
    - Population
    - Population growth
    - Personal income
    - Partisan voting index
  - Drop Honolulu HI and Anchorage AK and focus on 48 contiguous states
  - Temperature and Precipitation are collinear so use Köppen Aridity Index, which combines the two



Multilevel Model

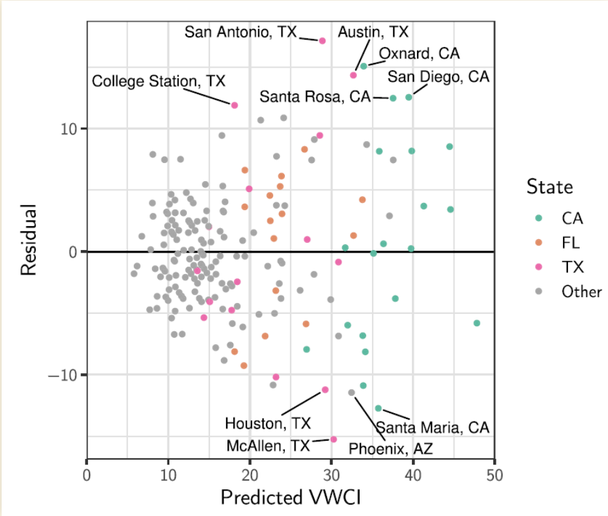
- Multilevel variable-intercept model:
  - Predict each city's score from city-level data and state-level data
    - Priors for intercepts are based on state-level data

$$V_i \sim \text{Binomial}(N_{\text{Actions}}, p_i)$$

$$\text{logit}(p_i) = \alpha_j + \sum_{k \in \text{city variables}} \beta_k x_{ik}$$

$$\alpha_j \sim \text{Normal}(\mu_j, \sigma_\alpha)$$

$$\mu_j = \alpha_0 + \sum_{k \in \text{state variables}} \gamma_k w_{jk}$$

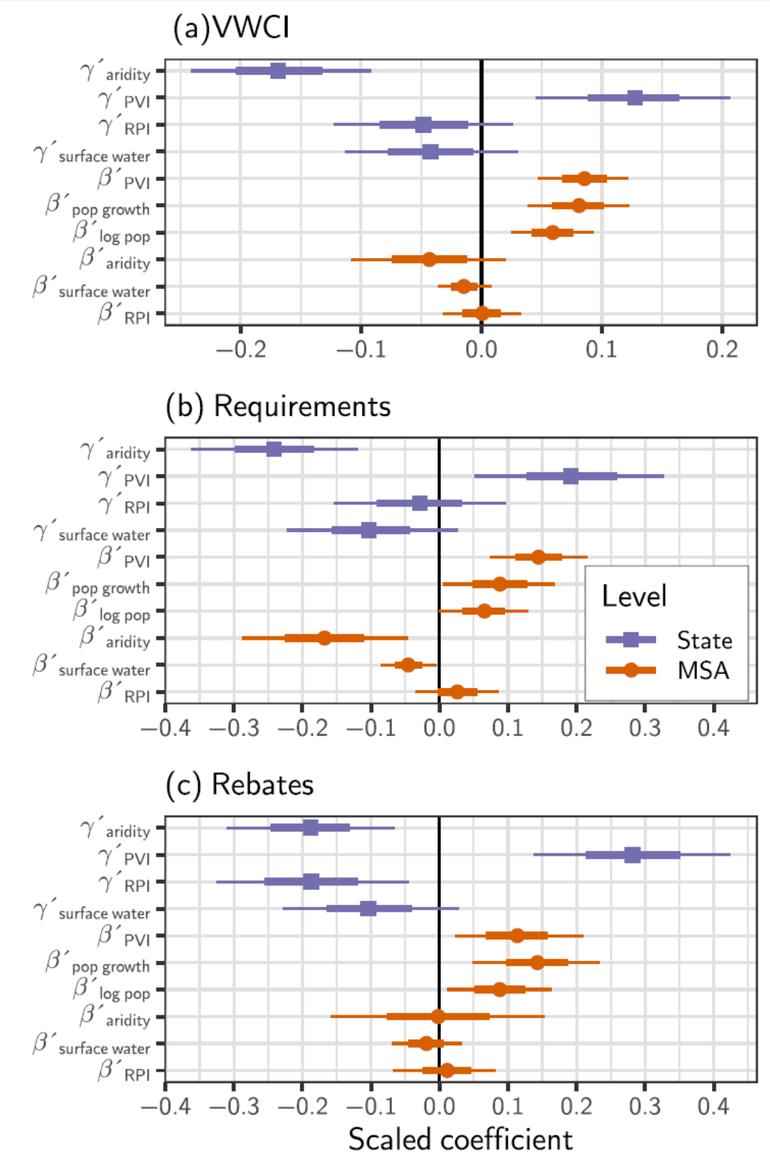


- Original residuals showed data were *overdispersed* (variance was too great for a Binomial), so we changed the model to use a *beta-binomial* distribution (see Chapter 12).

$$V_i \sim \text{beta-Binomial}(N_{\text{Actions}}, \phi p_i, \phi(1 - p_i))$$

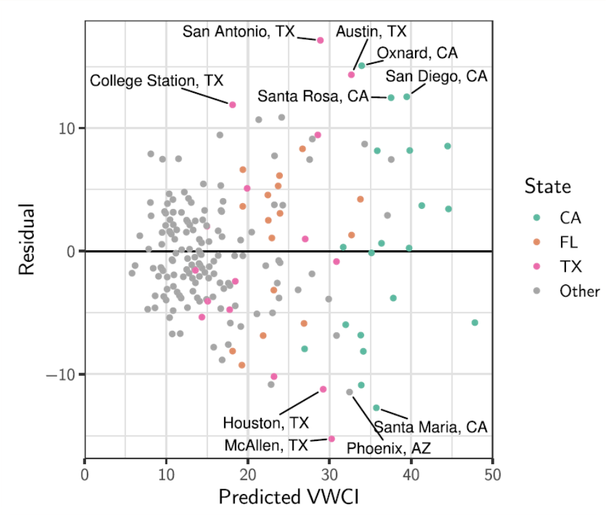
Results

- Most important state-level predictors:
  - Aridity
  - Partisan voting index (PVI)
- Most important city-level predictors:
  - PVI
  - Population growth
  - Log population
- City-level Aridity, surface water, and personal income don't matter after accounting for state-level effects



Checking model

- We did detailed interviews with water managers in San Antonio and Phoenix
  - San Antonio has a higher score than predictied.
    - Its Republican political leaning suggests low water conservation,
    - but the city doesn't have so much choice:
      - A lawsuit over endangered species led to federal requirements to conserve water
  - Phoelix has a lower score than predicted.
    - Central Arizona Project brings water from Colorado River
    - Reduces water stress on Phoenix



Cities With the 10 Largest Residuals From VWCI Regression

Rank	City	VWCI	predicted VWCI	residual
1	San Antonio, TX	46	28.9	17.1
2	McAllen, TX	15	30.2	−15.2
3	Oxnard, CA	49	33.9	15.1
4	Austin, TX	47	32.7	14.3
5	Santa Maria, CA	23	35.7	−12.7
6	San Diego, CA	52	39.5	12.5
7	Santa Rosa, CA	50	37.5	12.5
8	College Station, TX	30	18.1	11.9
9	Phoenix, AZ	21	32.4	−11.4
10	Houston, TX	18	29.2	−11.2